

BuildIMS — Presentation Cheat Sheet

Step-by-step demo guide · Keep this open during presentation

■ BEFORE YOU START — Run These First

Service	Command	Port
Backend	<code>cd E:\BuildIMS\backend && npm run dev</code>	4000
Frontend	<code>cd E:\BuildIMS\frontend && npm run dev</code>	3000
ML Server	<code>py -3 E:\BuildIMS\ml_server.py</code>	5001
ngrok	<code>ngrok http 3000 --host-header="localhost:3000"</code>	Public URL

■ Add ngrok URL to CORS in backend/src/app.js → *allowedOrigins*, then restart backend

→ Admin login: anaskhanbjr84@gmail.com / NewPass123

■ DEMO FLOW — Follow This Order

1. REGISTRATION + LOGIN

- Go to /register → Register a new student
- Check email inbox for verification link
- Click verify → Log in with new account
- Show DiceBear robot avatar auto-generated

2. FEED + LIVE ACTIVITY

- Go to / (Home Feed)
- Show real-time activity items
- Show Socket.io live updates (open 2 tabs)

3. PROJECT SHOWCASE

- Go to /showcase
- Click Submit Project (+Submit button)
- Fill title, description (100+ chars), program, semester
- Select tech tags, add GitHub URL
- Submit → goes to pending moderation
- Admin → Moderation → Approve it
- Refresh /showcase → project appears
- Vote on a project (▲ button)

4. DAILY CHALLENGES

- Go to /challenges
- Show today's challenge with completion counter
- Show weekly schedule (Mon=Code, Tue=Design...)
- Click Take Challenge → Submit a solution
- Show streak badges (Bronze 7d / Silver 30d / Gold 100d)

5. ML AUTO-GENERATION

- Go to /admin → Challenges tab
- Click ■ Auto-Generate with ML button
- Show generated challenge title in toast
- Explain: DistilGPT-2 trained on 1,408 LeetCode problems
- Loss dropped from 3.018 → 1.351
- Cron runs Mon-Fri at 8:50 AM PKT automatically

6. LEADERBOARD

- Go to /leaderboard
- Show 5 tabs: Overall, Streaks, Projects, Challenges, Programs
- Show program rankings (BSCS vs BSSE etc)

7. SKILLS GAP RADAR

- Go to /radar
- Show top in-demand skills for Pakistan
- Explain: JSearch API scrapes job postings weekly
- If empty: Admin → trigger scrape from browser console

8. IDEA GARAGE

- Go to /ideas
- Click Post Idea → fill title + description
- Show 21-day countdown starting immediately
- Explain: no proof in 21 days = "abandoned idea" on profile
- Show collaborate button on other ideas

9. SEMESTER SHOWDOWNS

- Go to /showdowns
- Show AI Warfare Showdown LIVE with countdown
- Click on it → show rules, description
- Click Submit Your Entry → fill and submit
- Admin → Showdowns → change status to Judging
- Show voting on submissions (▲ button)

10. ADMIN PANEL

- Go to /admin
- Dashboard: show MVP criteria progress bars
- Moderation: approve/reject pending projects
- Challenges: create + ML generate
- Showdowns: create + manage status
- Users: search, change role/status
- Digest: send weekly email now

11. USER PROFILE

- Go to /u/anas (your profile)
- Show streak, projects, badges
- Go to /settings → update avatar
- Upload photo or keep robot avatar

■ QUICK FIXES — If Something Breaks Live

Problem	Fix
Login says "Account locked"	<code>psql -U postgres -d buildims -c "UPDATE users SET failed_login_attempts=0, locked_until=NULL WHERE ..."</code>
Radar shows no data	Browser console: <code>fetch('/api/radar/scrape',{method:'POST',headers:{Authorization:`Bearer \${local...`</code>
ML generate fails	Check <code>py -3 E:\BuildIMS\ml_server.py</code> is running in separate PowerShell
Feed not loading	Check Redis (Memurai) is running: <code>Get-Service Memurai</code>
ngrok URL changed	Add new URL to <code>backend/src/app.js</code> <code>allowedOrigins</code> array, restart backend
Challenge already exists today	<code>psql: DELETE FROM challenges WHERE publish_date = CURRENT_DATE;</code> then retry ML generate
Project not showing after approve	Hard refresh browser (Ctrl+Shift+R)
Socket.io not connecting	Check backend console for Socket.io errors, restart backend

■ KEY URLS + CREDENTIALS

Page	URL	Notes
Home Feed	<code>localhost:3000/</code>	<i>Live activity feed</i>
Showcase	<code>localhost:3000/showcase</code>	<i>Project voting</i>
Challenges	<code>localhost:3000/challenges</code>	<i>Daily ML-generated tasks</i>
Leaderboard	<code>localhost:3000/leaderboard</code>	<i>5 tabs</i>
Skills Radar	<code>localhost:3000/radar</code>	<i>Job market data</i>
Idea Garage	<code>localhost:3000/ideas</code>	<i>21-day commitment board</i>
Showdowns	<code>localhost:3000/showdowns</code>	<i>48h hackathons</i>
Admin Panel	<code>localhost:3000/admin</code>	<i>Requires admin role</i>
Your Profile	<code>localhost:3000/u/anas</code>	<i>Public profile</i>
Backend API	<code>localhost:4000/api</code>	<i>REST endpoints</i>
ML Server	<code>localhost:5001/health</code>	<i>Flask + DistilGPT-2</i>
HuggingFace	<code>huggingface.co/anascs12/buildims-challenge-generator</code>	<i>Generated model</i>

Admin Email	<code>anaskhanbjr84@gmail.com</code>	Admin Password	<code>NewPass123</code>
DB User	<code>postgres</code>	DB Name	<code>buildims</code>
DB Password	<code>buildims123</code>	Redis	<code>No password (Memurai)</code>
HF Token	<code>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</code>	HF User	<code>anascs12</code>

■ ML MODEL — What to Say to Teacher

What we did:

- Downloaded LeetCode dataset from Kaggle (1,825 problems)
- Cleaned to 1,408 free, non-null problems
- Formatted as structured text with TITLE/TYPE/LEVEL/DESCRIPTION
- Fine-tuned DistilGPT-2 (82M parameters) on Google Colab free T4 GPU
- 4 epochs, 1,013 training blocks, ~3 hours training time
- Loss: 3.018 → 1.351 (good convergence for small model)

Architecture:

- Base model: distilgpt2 (HuggingFace)
- Custom tokens: `<|challenge|>` and `<|endchallenge|>`
- Vocab expanded: 50,257 → 50,260 tokens
- Inference: local Flask API on port 5001
- Integration: Node.js backend calls `localhost:5001/generate`
- Auto-cron: Mon-Fri 8:50 AM PKT generates challenge if none exists